A Project Report on

GeoValuator: An Integrated Geospatial System for Land Mapping and Property Analysis

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We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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ABSTRACT

The last few years have seen considerable changes in land administration and real estate owing to the use of data-oriented technologies. The global demand for land is on the rise due to rapid urbanization, population growth, and shifting market dynamics. Thus, there is a desperate need to make great advances in land valuation and mapping in a reliable, transparent manner. While time-tested, conventional property assessment practices prove inadequate to address the intricacies and vagaries of modern real estate markets and legal regulations.

Machine learning, geospatial mapping, and data analytics begin to look like potent allies for stakeholders, allowing them to make informed decisions concerning property valuation and associated risks. These technologies permit intelligent land administration systems to forecast prices, evaluate legal risks, and predict optimal land usage through large datasets. Supported by historical land records and real-time geospatial inputs, efficient valuation systems can increase accuracy, minimizing disputes, and improving property management in real time subsequently.

The current project reviews the appreciable link of artificial intelligence in geospatial systems, which has the potential to change the conventional decision-making processes in real estate investment and land mapping. With the digital transformation era, property administration is more and more adopting data-driven approaches to improve transparency and accuracy. This project introduces GeoValuator, a complete machine learning-based geospatial framework for land mapping, price forecasting, and legal risk assessment as support for investors, landowners, and urban planners to make wise decisions.

Keywords: Geospatial mapping, land valuation, artificial intelligence, real estate analytics, price forecasting, legal risk assessment, decision support system, predictive land usage, smart property management, integrated geospatial system.

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1. INTRODUCTION

In the wake of rapid urbanization, rising property values, and the increasing complexity of real estate transactions, the land management sector is facing significant challenges. Traditional methods of land monitoring and valuation are often plagued by a lack of transparency, manual inefficiencies, and a heavy reliance on local intermediaries rather than data-driven insights. As a result, property owners—especially those residing in different cities or countries—frequently face the risk of land encroachment (Kabja) and struggle to obtain accurate, market-reflective valuations for their assets.

The integration of modern technologies such as Machine Learning, Computer Vision, and Remote Sensing has paved the way for the development of intelligent land analytics tools like GeoValuator. This system leverages high-fidelity historical datasets and satellite imagery to provide a dual-layered solution: precise market price prediction using the XGBoost regressor and proactive encroachment detection through OpenCV-based image analysis. By adopting this automated platform, stakeholders can transition from reactive monitoring to a proactive "Land Health" approach, significantly reducing legal risks, ensuring fair market pricing, and enhancing the overall security of real estate investments.

1.1 BACKGROUND

In India, land is often seen as the most secure and significant long-term investment a person can make. Whether it is inherited property or a plot purchased with life savings, land represents more than just financial value; it is a core part of a family’s security. However, managing these assets has historically been a manual and stressful task. Property owners frequently lack the tools to verify their land's safety or its true market worth, leading to a "transparency gap" where they must rely on local intermediaries or expensive physical surveys. This lack of data-driven insight often results in legal disputes, loss of property through encroachment (Kabja), and unfair pricing during transactions.

The recent explosion in Machine Learning and the availability of high-resolution satellite data have opened a new door for property management. While moving toward an era of "intelligent monitoring," where the physical state of a plot can be tracked from anywhere in the world. Modern predictive models can now analyze dozens of variables—from soil quality and local infrastructure to historical market trends—to calculate a property's value with a precision that was impossible just a few years ago. Similarly, computer vision algorithms can now act as a digital "eye," instantly flagging any unauthorized physical changes on a site.

Data is the fuel for this transformation. By processing diverse points such as geographical coordinates, land use zones, connectivity scores, and satellite imagery, by this can build a comprehensive "Land Health Index." These systems allow for a proactive approach to land security, where risks are identified before they become legal nightmares.

This project, GeoValuator, was developed to bridge the gap between traditional land ownership and modern technology. It combines two essential pillars of land security—accurate valuation and risk monitoring—into a single, user-friendly web portal. Under the hood, the system uses the XGBoost regressor for high-fidelity price forecasting and OpenCV-based image analysis to detect encroachment in real-time satellite photos. Ultimately, GeoValuator empowers property owners with the scientific clarity they need to protect their most valuable assets in an increasingly digital world.

1.2 MOTIVATION

Real estate is far more than just a sector; it is the cornerstone of long-term wealth and the primary source of financial security for millions of families. However, the current methods used to manage land are often outdated, riddled with transparency issues, and heavily reliant on manual effort. As the demand for property grows and urban landscapes shift, there is an urgent need for smarter, data-driven platforms that can protect assets and provide clear financial insights. This project is motivated by the desire to move away from guesswork and replace it with a high-security, AI-driven "digital shield" for landowners. The core motivation stems from the following critical challenges:

1.2.1 Inefficiency in Physical Land Monitoring

Traditional land monitoring is a slow and costly process that typically requires the owner's physical presence. For those living far from their property—especially NRIs or people residing in different cities—frequent visits are nearly impossible. This creates a "monitoring blind spot" where issues are often discovered only after they have become legal nightmares. An automated satellite-based system can remove this inefficiency, allowing for constant vigilance from anywhere in the world.

1.2.2 Fragmented and Opaque Valuation Systems

Currently, property valuation is a "black box" controlled by local brokers and intermediaries. This often leaves owners in the dark about the true market value of their land, as generic pricing figures rarely account for micro-factors like soil quality, connectivity, or regional infrastructure. By using scientific data, this can move away from these opaque practices and toward a system where every price is backed by verifiable data.

1.2.3 Rising Threats of Land Encroachment (Kabja)

Illegal encroachment, or "Kabja," remains one of the most significant risks for property owners in developing regions. Despite having legal documents, the physical security of a plot is often compromised when it is left unattended. Motivated by the need for early warning, this project uses computer vision to detect unauthorized boundary changes, providing a level of protection that manual records simply cannot offer.

1.2.4 Need for Hyper-Localized Property Insights

Broad real estate trends are often useless to the individual plot owner because land value changes significantly from one acre to the next. Whether it's the proximity to a beach, distance from a highway, or the specific "Zone Type," these localized factors are critical. A data-driven system can process these unique variables to offer a "hyper-local" perspective that is far more accurate than any generalized market report.

1.2.5 Harnessing AI for Real Estate Security

The recent breakthroughs in machine learning and image processing have opened a massive opportunity to build predictive systems for personal assets. The ability of XGBoost algorithms to handle complex market data and OpenCV to analyze satellite imagery presents a powerful path for automating land safety. This project is inspired by the potential to put these sophisticated technologies into the hands of everyday landowners, making property management safer and more scientific.

1.3 PROBLEM STATEMENT

Land ownership remains a cornerstone of personal wealth and economic stability, particularly in regions where property serves as a primary financial asset. Despite the rapid advancement of digital tools, many landowners continue to manage their property based on intuition, hearsay from local intermediaries, or outdated manual records that fail to reflect the actual state of the land. This often results in inaccurate property valuations, a lack of transparency in transactions, and a constant, unresolved anxiety regarding illegal encroachment (Kabja). The absence of scientific, data-driven tools accessible to individual property owners contributes to significant challenges in maintaining land security and asset value.

One major problem is the absence of an integrated platform that offers both accurate land valuation and real-time risk monitoring based on precise geographical and market data. While some platforms focus on listing properties, they do not provide a scientific way to estimate fair market value based on infrastructure and soil parameters. Conversely, security systems may offer physical monitoring but fail to account for the financial "health" and price trends of the land. Additionally, external factors like land use zoning, distance to key amenities, and local legal disputes vary widely across regions, making it difficult for owners—especially those living far away—to maintain a clear and up-to-date picture of their assets.

The increasing complexity of real estate markets and the growing threat of unauthorized construction further exacerbate these issues, demanding the need for intelligent and adaptive monitoring systems. Landowners require decision-support tools that are data-driven, localized, and capable of detecting patterns in physical land

changes and market fluctuations. The challenge lies in developing a solution that not only provides an accurate price forecast but also monitors for physical boundary violations using historical and real-time imagery.

GeoValuator addresses these challenges by proposing a dual-engine solution—a Price Prediction System that uses infrastructure and geographical features to calculate property value, and an Encroachment Detection System that uses satellite imagery and computer vision to flag unauthorized physical changes. This unified, AI-powered system bridges the gap between valuation and security, enabling property owners to make informed, data-driven decisions while protecting their assets from anywhere in the world.

1.4 OBJECTIVE OF PROJECT

The primary objective of this project is to develop a robust and intelligent land analytics and security system that integrates both automated valuation and encroachment detection based on geographical, infrastructural, and satellite parameters. This system aims to assist property owners, real estate investors, and administrators in accurately estimating land value and monitoring physical changes remotely, thereby promoting transparency in the property market and enhancing asset protection.

1.4.1 Development of a Precise Land Valuation Engine

The first key objective is to build a high-performance machine learning model that predicts the fair market value of property based on multi-dimensional data. This includes analyzing geographical coordinates, total land area, road connectivity, and regional proximity to critical infrastructure. The goal is to provide a scientific alternative to traditional broker-led pricing, ensuring that every valuation is backed by verifiable data.

1.4.2 Implementation of a Satellite-based Encroachment Detection System

Another core objective is to design an automated surveillance system using computer vision techniques. By comparing historical and current satellite imagery, the system will be capable of identifying unauthorized construction or boundary violations. This objective focuses on providing a "digital watchdog" for property owners, allowing them to maintain vigilance over their assets from anywhere in the world.

1.4.3 Comparative Evaluation of Machine Learning Models

An essential part of this research involves benchmarking multiple machine learning algorithms to ensure the highest possible accuracy. This includes experimenting with models such as Linear Regression, Random Forest, and XGBoost. These models are evaluated against key performance metrics, including R-squared (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), to identify the most reliable engine for real-world deployment.

1.4.4 Integration into a Unified Web Dashboard

The final objective is to ensure the practical utility of the project by integrating these AI engines into a seamless, user-friendly web interface. Built using modern stacks like React, Node.js, and FastAPI, this dashboard will allow users to input property details or view satellite maps to receive real-time valuations and encroachment alerts without requiring any technical expertise.

1.5 SIGNIFICANCE

The integration of Machine Learning and remote sensing into land management has opened new avenues for enhancing property security and asset valuation processes. This project holds significant value in the context of modern real estate, where data-driven insights are becoming increasingly vital for financial transparency, legal protection, and efficient resource allocation.

One of the most important contributions of this project is its ability to help property owners determine the true, data-backed market value of their land. In traditional real estate, pricing is often opaque and controlled by intermediaries, which may not always reflect the actual value. By using a valuation model trained on historical and geographical data—including distance to infrastructure, soil quality, and connectivity—this system provides personalized and data-informed pricing. This leads to fairer transactions, improved financial transparency, and a minimized risk of being misled by inconsistent market trends.

In addition to predicting property value, the system also addresses the anxiety associated with remote land monitoring. The automated encroachment detection feature allows owners—especially those living in different cities or abroad—to maintain constant vigilance over their assets. Accurate alerts regarding physical changes or unauthorized construction aid in timely legal intervention and long-term planning. For government bodies and urban planners, these insights can also assist in monitoring unauthorized land use and managing regional development more effectively.

Moreover, this dual-engine project helps bridge the gap between complex spatial data and actionable intelligence. By employing a user-friendly web interface, even non-technical users can benefit from advanced predictive models without needing expertise in machine learning or satellite analysis. This democratization of technology is particularly valuable for individual landowners who were previously excluded from the sophisticated tools used only by large-scale developers.

Lastly, the project promotes the use of intelligent monitoring frameworks that encourage further research in smart land management. Its adaptability allows for future scaling, such as integration with mobile applications, government land record databases (Blockchain), or street-view cameras, expanding its impact and utility even further.

2. LITERATURE SURVEY

The field of land valuation and land management has undergone a significant transformation with the advancement of geospatial technologies, machine learning, and digital information systems. Traditionally, land valuation relied on manual inspection, expert judgment, and limited comparative market analysis. These approaches often lacked consistency, scalability, and transparency, leading to inaccurate pricing, disputes over ownership, and inefficiencies in land governance systems. As a result, researchers and practitioners have increasingly explored data-driven and technology-enabled approaches to improve land valuation accuracy and management efficiency.

One of the earliest developments in property valuation was the use of hedonic pricing models, which estimate land value based on various attributes such as location, accessibility, and environmental conditions. These models established that geographic proximity to infrastructure such as roads, schools, hospitals, and commercial centers plays a critical role in determining land prices. Further research introduced the concept of spatial autocorrelation, which demonstrated that nearby land parcels tend to have similar valuation patterns. This insight laid the foundation for integrating spatial data into valuation systems.

With the rise of machine learning, researchers began applying regression-based models for property price prediction. Techniques such as Linear Regression, Decision Trees, Random Forest, and Support Vector Machines have been widely studied for estimating land and real estate prices. Among these, ensemble learning methods like Random Forest and Gradient Boosting have shown superior performance due to their ability to model complex, non-linear relationships between input variables. These models effectively capture interactions between geographic, economic, and infrastructural features, making them suitable for real-world land valuation scenarios.

Geographic Information Systems (GIS) have also emerged as a powerful tool for land management and mapping. GIS enables the visualization and analysis of spatial data, allowing users to interactively explore land parcels based on their

geographic coordinates. Modern land information systems incorporate GIS to map land boundaries, assign unique identifiers such as survey numbers, and link spatial data with ownership records. This integration significantly improves transparency, reduces manual errors, and enhances decision-making.

Recent studies have further explored the use of remote sensing and satellite imagery for monitoring land usage and detecting unauthorized activities. Image processing techniques such as change detection and pixel-level analysis have been successfully applied to identify encroachments and structural changes in land parcels. These technologies enable continuous monitoring of land assets, providing real-time alerts to landowners and authorities.

Another important area of research is digital ownership verification and land registry systems. Secure frameworks have been proposed to link land parcels with verified owner identities using unique identifiers, ensuring authenticity and preventing fraudulent transactions. These systems also maintain audit trails of ownership transfers, improving accountability and legal compliance.

In addition to valuation and ownership management, researchers have introduced risk assessment models to evaluate land vulnerability. These models consider factors such as dispute history, ownership verification status, and physical conditions to classify land parcels into different risk categories. Such frameworks enable proactive identification of high-risk properties and support better land governance.

Despite these advancements, most existing systems focus on individual aspects such as valuation, mapping, or ownership verification, without integrating them into a unified platform. Moreover, many models rely on global datasets that may not accurately represent local conditions, resulting in reduced prediction accuracy. There is also a lack of systems that combine predictive analytics with interactive visualization, limiting user understanding and usability.

The GeoValuator system addresses these gaps by integrating machine learning-based price prediction, GIS-based visualization, digital ownership verification, and risk assessment mechanisms into a single platform. By leveraging locally collected data and combining multiple technologies, the system provides a

comprehensive solution for modern land management challenges, ensuring accuracy, transparency, and security.

2.1 EXISTING SYSTEM

In the current real estate and land management landscape, several platforms and tools have been developed to aid property owners in searching for records and estimating market prices. However, most of these systems either focus on static listings based on broad urban zones or provide manual valuation guidelines that lack personalization and real-time adaptability. The existing systems often fail to utilize the full potential of modern data science and machine learning techniques to provide dynamic, accurate, and security-focused land assessments.

Traditional land valuation methods rely heavily on regional "base rates," local broker hearsay, or manual appraisal by intermediaries. These methods, while common, do not account for the specific variations in soil properties, road connectivity scores, or micro-location changes that can significantly impact a plot's actual market performance. As a result, the risk of financial loss due to inappropriate pricing or lack of transparency remains high.

When it comes to security and monitoring, most conventional approaches are reactive, relying on physical visits or manual surveys. These methods typically lack the ability to provide constant oversight for owners who reside far from their property. In many cases, encroachment (Kabja) or unauthorized construction is only detected after legal complications have matured, by which time the resolution becomes both difficult and expensive.

With the rise of Machine Learning technologies, multiple systems for real estate price prediction were designed. Most of these systems were built using basic linear models or simple Decision Trees that rely on a limited set of parameters like location and square footage, often ignoring more complex features like land-use zoning or risk factors.

Furthermore, many existing GIS and satellite monitoring tools are high-end professional applications with limited accessibility for individual landowners. They

often require specialized technical skills or expensive subscriptions, which may not be feasible for a common person looking for simple land security.

Additionally, there is a significant lack of integration between valuation and m onitoring systems. This fragmentation forces property owners to rely on separate platforms for financial insights and physical security alerts, leading to inefficiencies and increased uncertainty. The need for an integrated, intelligent, and user-friendly system that simultaneously predicts fair market price and detects encroachment, based on real-time inputs and advanced machine learning techniques, forms the core motivation behind the development of the proposed GeoValuator system.

2.2 Disadvantages of Existing System

The existing land management and valuation systems have several limitations that reduce their effectiveness and reliability.

Disadvantages:

- Land price estimation is done manually, which leads to inaccurate pricing.
- Users must visit multiple websites to get land ownership, mapping, and price details.
- Existing systems do not provide real-time price prediction.
- Most systems do not include legal risk analysis or dispute detection.
- GIS systems require technical knowledge and are not user-friendly.
- No integration between land mapping, ownership verification, and price estimation.
- Lack of transparency increases the risk of fraud and illegal land occupation.
- Traditional methods are time-consuming and inefficient.

Because of these disadvantages, there is a need for an intelligent and integrated system that can combine mapping, ownership verification, and price estimation into one platform.

2.3 PROPOSED SYSTEM

The proposed system, GeoValuator, is an intelligent and integrated geospatial platform designed to provide accurate land price estimation, ownership verification, and risk analysis using modern data-driven techniques. The system combines Geographic Information Systems (GIS), machine learning algorithms, and web technologies to create a unified framework for efficient land management and decision support.

The architecture of GeoValuator is designed as a multi-layered system, consisting of the Data Layer, Intelligence Layer, and Application Layer. The Data Layer is responsible for storing structured land records, including geographic coordinates, land attributes, ownership details, and pricing information. A NoSQL database such as MongoDB is used to efficiently handle structured and semi-structured data, allowing flexible storage and retrieval of land information.

The Intelligence Layer forms the core of the system, where machine learning models are used to predict land prices and assess legal risks. Regression-based algorithms analyze multiple input features such as land area, location, connectivity, infrastructure proximity, zoning classification, and historical price trends to generate accurate price predictions. Additionally, a risk evaluation mechanism, known as the Kabja Risk Engine, analyzes legal and ownership attributes to classify land parcels into different risk categories.

The Application Layer provides a user-friendly web interface that enables users to interact with the system. The frontend, developed using React, allows users to input land details and visualize results on an interactive map. The backend, implemented using Node.js and Express, handles data processing and communication between the frontend and the AI engine. The AI engine, built using FastAPI, processes input data and returns predictions in real time.

A key feature of the system is its geospatial visualization capability, which allows users to view land parcels on a digital map using their geographic coordinates. Each parcel is represented by a marker that displays information such as predicted

price, ownership details, and risk classification. This visual representation enhances user understanding and supports informed decision-making.

The system also incorporates advanced features such as the Land Health Index (LHI), which provides an overall assessment of land quality and risk, and a satellite-based monitoring mechanism that detects unauthorized changes using image processing techniques. Furthermore, an AI-powered conversational assistant is integrated to enable natural language queries, making the system accessible to non-technical users.

By integrating multiple functionalities into a single platform, GeoValuator overcomes the limitations of traditional systems and provides a comprehensive solution for land valuation and management.

2.4 Advantages of Proposed System

The proposed GeoValuator system provides several advantages compared to existing systems.

Advantages:

- Provides accurate land price prediction using machine learning.
- Displays land parcels on an interactive GIS map.
- Integrates land ownership, mapping, and price estimation in one system.
- Provides legal risk analysis and dispute detection.
- Reduces manual work and saves time.
- Improves transparency in land valuation.
- Helps buyers and sellers make better decisions.
- Provides real-time land information and price estimation.
- User-friendly interface that can be used by non-technical users.

Thus, the proposed system provides a complete and intelligent solution for land valuation, ownership verification, and land monitoring.

3. ANALYSIS

The analysis phase is an important stage in the development of the GeoValuator system. In this phase, the system requirements, user requirements, hardware and software requirements are identified and analyzed. The main objective of system analysis is to understand the functionality of the system, identify user needs, and determine the technical requirements required to develop the system.

The GeoValuator system is designed to provide land price prediction, ownership verification, and risk assessment using machine learning and geospatial technologies. The analysis phase focuses on understanding the data requirements, processing methods, system architecture, and user interaction with the system. This phase helps in designing an efficient and reliable system architecture for implementation.

3.1 Software Requirement Specification

Software Requirement Specification (SRS) is a document that describes the functional and non-functional requirements of the system. It provides a detailed description of system functionality, user interactions, system constraints, and performance requirements. The SRS document helps developers understand what the system should do and how the system should perform.

In the GeoValuator system, the SRS includes user requirements, software requirements, and hardware requirements. The system is designed to accept land-related inputs, process the data using machine learning algorithms, and generate outputs such as land price prediction, risk assessment, and geospatial visualization. The SRS also describes the technologies used for system development and the overall system architecture.

3.1.1 User Requirement

User requirements describe the services and functionalities that the system should provide to users. The GeoValuator system is designed for landowners, buyers,

government authorities, and administrators who need information about land price, ownership, and risk status.

The user should be able to enter land details such as survey number, land area, location coordinates, zoning type, and infrastructure proximity through a user-friendly interface. The system should display the predicted land price, risk status, and land details on an interactive map. The user should also be able to search land records and view ownership information.

The system should provide accurate results, easy navigation, and fast response time so that users can make informed decisions regarding land valuation and management.

3.1.2 Software Requirements

Software requirements specify the software tools, programming languages, frameworks, and libraries required to develop the system. The GeoValuator system is developed using modern web technologies and machine learning frameworks.

Software Requirements:

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Node.js and Express.js
- Database: MongoDB
- Machine Learning: Python, FastAPI
- Mapping: Leaflet.js and OpenStreetMap
- Development Tools: Visual Studio Code
- API Testing: Postman
- Operating System: Windows / Linux / macOS
- Browser: Google Chrome / Microsoft Edge

These software tools are used to develop the web application, process land data, train machine learning models, and display geospatial data on the map.

3.1.2.1 Importing Libraries

A comprehensive set of libraries and frameworks was integrated to support geospatial data analysis, AI model development, and full-stack web integration:

i. Pandas and NumPy: Utilized for high-performance data manipulation, handling land datasets, and numerical computations.

ii. Matplotlib and Seaborn: Employed for generating critical data visualizations, including feature correlation heatmaps and price distribution histograms.

iii. Scikit-learn (sklearn): Provided the foundational machine learning utilities for data preprocessing (Label Encoding), model evaluation (R^2 Score, MAE), and cross-validation.

iv. XGBoost: A high-performance gradient boosting framework used as the core engine for land price estimation due to its handling of complex non-linear relationships.

v. Joblib/Pickle: Used for serializing the trained XGBoost models and encoders to ensure consistent inference during deployment.

vi. Node.js & React (Vite): Developed the modern web interface, replacing legacy frameworks with a responsive React frontend and a robust Node.js backend to handle real-time geospatial queries.

3.1.3 Hardware Requirements

Hardware requirements specify the physical components required to run the system. The GeoValuator system does not require special hardware components and can run on a standard computer system.

Hardware Requirements:

- Processor: Intel i3 or above
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: Minimum 10 GB free space

- Internet Connection: Required
- GPS-enabled device (optional for location detection)

These hardware components are sufficient for running the web application, database, and machine learning model.

3.2 CONTENT DIAGRAM OF PROJECT

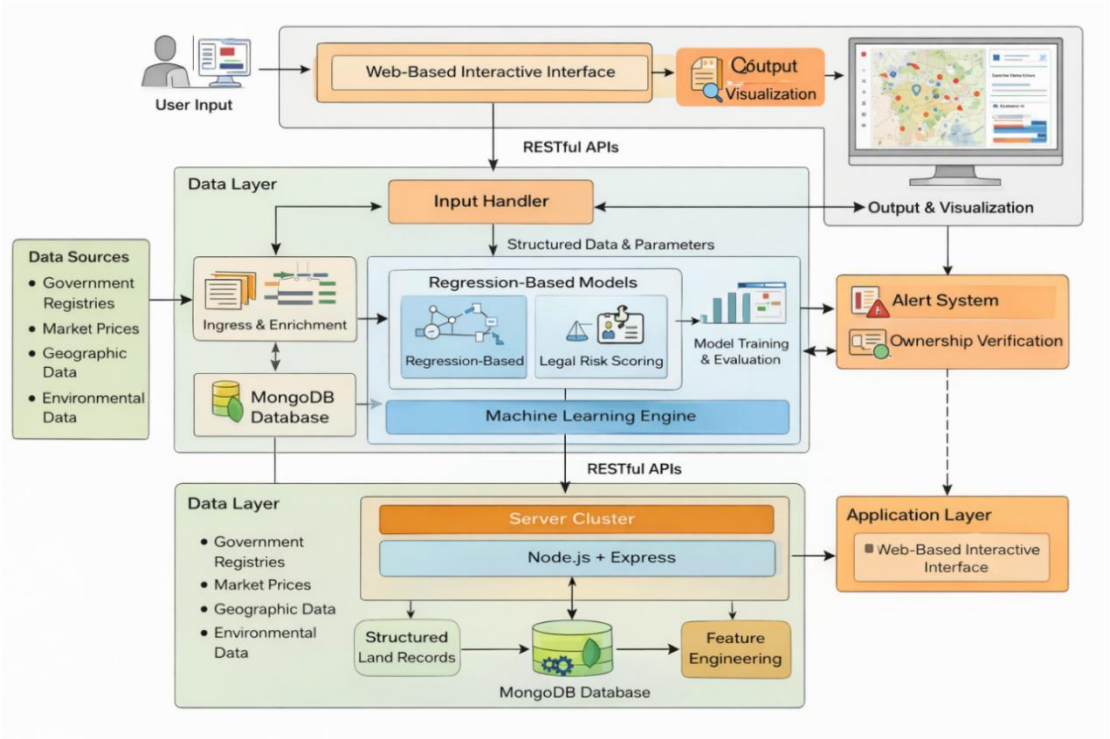

Fig.3.1 GeoValuator Architecture

The content diagram of the GeoValuator system represents the overall architecture and data flow between different components of the system. The system is divided into multiple layers including the user interface layer, data layer, machine learning layer, server layer, and application layer.

Initially, the user provides land-related inputs through the web-based interactive interface. The input data is sent to the backend server through RESTful APIs. The backend server processes the input data and stores it in the MongoDB database. The structured land data is then sent to the machine learning engine, where regression-based models are used to predict land price and legal risk score.

The machine learning engine performs model training, evaluation, and prediction using structured land parameters such as location, land area, zoning type, and infrastructure proximity. The predicted results are sent back to the server through REST APIs and stored in the database.

The output is then sent to the output visualization module, where the results are displayed on an interactive map interface. The system also includes an alert system and ownership verification module to provide security and risk alerts to users.

Thus, the content diagram explains the overall workflow of the GeoValuator system and shows how data flows between different modules such as frontend, backend, database, and machine learning engine.

4. DESIGN

The system design encapsulates the structural and behavioral architecture of the proposed GeoValuator land analytics and monitoring framework. This design ensures that the platform is modular, scalable, and user-friendly, allowing it to handle complex AI workloads in real-time. The system is divided into several logical layers: the Frontend Interface Layer, the Backend Orchestration Layer, and the AI Processing Engine. These layers work cohesively to deliver accurate valuation reports and automated encroachment alerts based on user-provided geographical and satellite data.

4.1 SYSTEM WORKFLOW

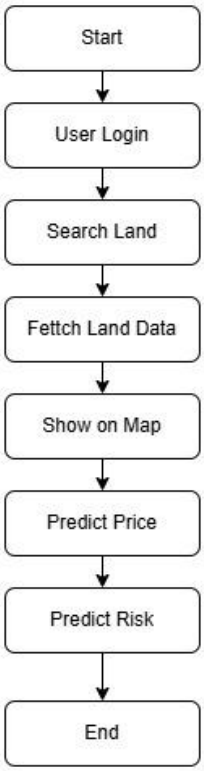

Fig. 4.1 GeoValuator Workflow

The system workflow describes the step-by-step operation of the GeoValuator system from user input to final output generation. The workflow begins when the user logs into the system and searches for land records using survey number or owner name. The system retrieves land data from the database and displays the land parcel on the interactive map using geographic coordinates. After retrieving land details, the

system performs land price prediction using the machine learning model and calculates the legal risk score using the risk analysis module. The predicted price and risk score are then displayed to the user along with land details. The system workflow ends after displaying the results to the user. The workflow of the system is illustrated in Fig. 4.1. This workflow ensures that the system processes user requests efficiently and provides real-time land analysis results.

4.2 CLASS DIAGRAM

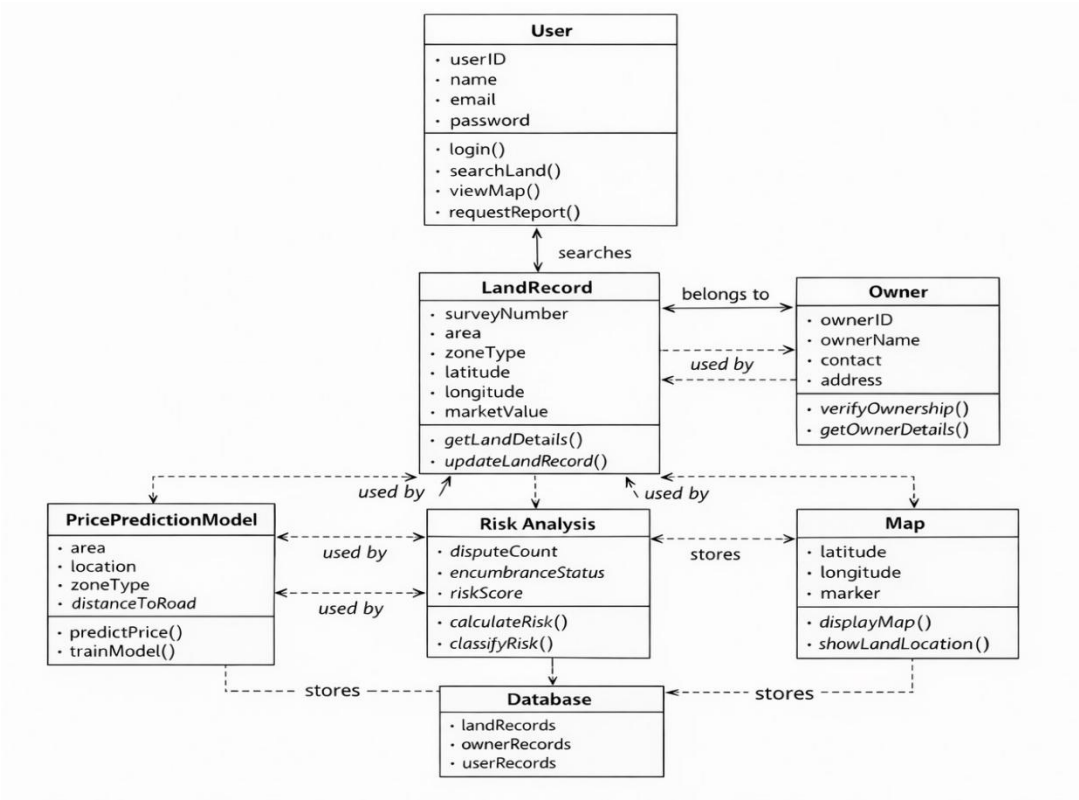

Fig. 4.2 GeoValuator Class Diagram

The class diagram represents the object-oriented structure of the GeoValuator system. The system consists of several classes including User, LandRecord, Owner, PricePredictionModel, RiskAnalysis, Map, and Database. Each class represents a major entity in the system and contains attributes and methods related to that entity. The User class handles user operations such as login, land search, map viewing, and report generation. The LandRecord class stores land details such as survey number, area, zone type, and geographic coordinates. The Owner class stores ownership details and verification information. The PricePredictionModel class is responsible for

predicting land price based on input features such as area, location, and zone type.

The RiskAnalysis class calculates the legal risk score based on dispute count and encumbrance status. The Map class displays land parcels on the map using geographic coordinates. The Database class stores land records, owner records, and user records. The relationships between these classes show how data flows between system components. The class diagram of the system is shown in Fig. 4.2.

4.3 USE CASE DIAGRAM

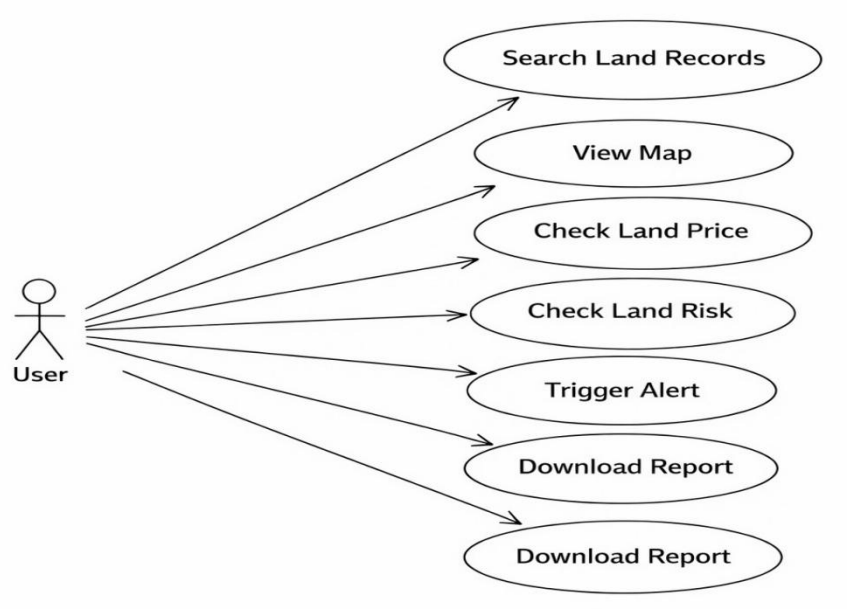

Fig. 4.3 GeoValuator User's Use Case Diagram

The use case diagram represents the interaction between the user and the GeoValuator system. The main actor in the system is the user who interacts with the system to perform various operations. The user can search land records, view land parcels on the map, check land price, check legal risk, trigger alerts, and download land reports. These use cases represent the major functionalities provided by the system. The use case diagram helps to understand the functional requirements of the system and how users interact with the system. The use case diagram of the system is shown in Fig. 4.3.

4.4 DESIGN ADVANTAGES

The system design provides several advantages such as modular architecture, real-time data processing, and integration of geospatial mapping with machine learning analytics. The modular architecture allows individual modules such as mapping, price prediction, and risk analysis to be updated independently without affecting the entire system. The use of a web-based interface makes the system accessible from any location. The integration of machine learning with geospatial data improves the accuracy of land price estimation and risk prediction. The system also improves transparency and reduces manual work in land verification and price estimation.

5. IMPLEMENTATION & RESULTS

5.1 METHOD OF IMPLEMENTATION

The GeoValuator system is implemented as a web-based application that integrates machine learning models, geospatial visualization, and a database management system to provide land price prediction and risk assessment. The system is developed using a multi-layer architecture consisting of frontend, backend, database, and machine learning modules.

The frontend of the system is developed using ReactJS, which provides a user interface for entering land details and viewing results on a map. The backend is developed using Node.js and Express.js, which handle API requests, data processing, and communication between the frontend and the machine learning module. The database used is MongoDB, which stores land records, owner details, and prediction results.

The machine learning module is implemented using Python and FastAPI. The machine learning models are trained using historical land data and used to predict land price and risk score. The system workflow includes data collection, data preprocessing, model training, prediction, and result visualization. The predicted results are displayed on an interactive map interface along with risk classification and land details.

5.1.1 Implementation of Algorithms

The GeoValuator system implements machine learning algorithms to predict land prices and evaluate legal risk associated with land parcels. Since land price prediction is a continuous value prediction problem, regression algorithms are used in the system. The implementation includes Linear Regression, Decision Tree Regressor, Random Forest Regressor, and XGBoost Regressor. These algorithms are trained using historical land data containing features such as land area, geographic

coordinates, zoning classification, road connectivity, infrastructure proximity, and legal attributes.

Linear Regression is implemented as a baseline model to establish a relationship between input features and land price. The model assumes a linear relationship between independent variables such as land area, distance to town, and zone type, and the dependent variable, which is the land price. This model helps in understanding the influence of each feature on land price and provides a simple prediction model.

The Decision Tree Regressor is implemented to capture non-linear relationships between land attributes and land price. The model divides the dataset into smaller subsets based on feature conditions such as zone type, distance to highway, and land area. This allows the system to model complex relationships between land features and price variations.

The Random Forest Regressor is implemented as an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. The model generates multiple decision trees using random subsets of data and features, and the final prediction is calculated as the average of predictions from all trees. This algorithm improves the stability and accuracy of land price prediction in the GeoValuator system.

The XGBoost Regressor is implemented as an advanced boosting algorithm that builds decision trees sequentially, where each new tree corrects the errors of the previous tree. This algorithm provides high accuracy and performs well with structured land datasets containing multiple features. Due to its high performance, XGBoost is used as the primary model for land price prediction in the system.

In addition to land price prediction, the system implements a legal risk assessment algorithm to evaluate the risk associated with land parcels. The risk score is calculated based on parameters such as dispute count, encumbrance status, ownership verification, and boundary status. Based on the calculated score, the land is classified into low risk, moderate risk, or high risk categories. This risk classification helps users identify legally sensitive land parcels.

The machine learning models are trained using the Bapatla land dataset and evaluated using performance metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2) score. The model with the best performance is selected and deployed in the system for real-time land price prediction. The trained model is integrated into the web application using FastAPI, and predictions are displayed to the user through the web interface along with geospatial visualization on the map.

Table 5.1 Algorithms Used in GeoValuator

Algorithm	**Purpose**
Linear Regression	Basic price prediction
Decision Tree	Non-linear price prediction
Random Forest	Accurate price prediction
XGBoost	High accuracy prediction
Risk Scoring Algorithm	Legal risk calculation

5.1.2 Output Screens

The user interface is designed using a modern web stack featuring a React frontend and a Node.js backend, integrated with a FastAPI based AI engine. This is the primary landing page of the application that appears when the services are initiated from the terminal and the application is accessed through a web browser (typically at http://localhost:5173).

The index page of the GeoValuator system is shown below. Fig. 7.1 illustrates the homepage design, which features a professional government-themed interface with two core interaction pathways: "Access Dashboard" for secure administrative and user functions, and "Public Map" for transparent access to the National Land Intelligence & Valuation Grid.

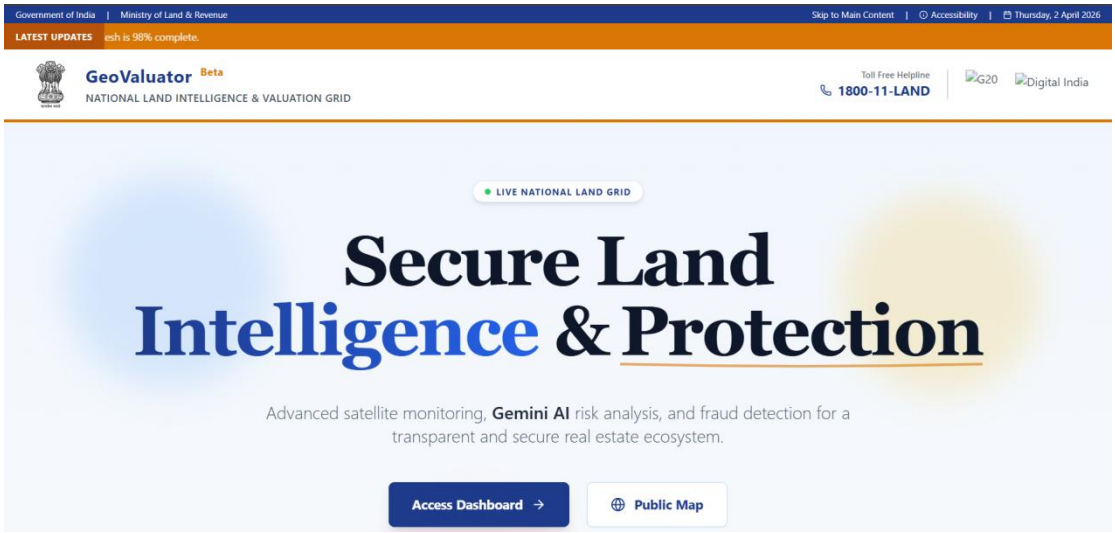

Fig. 5.1 GeoValuator Landing Page Interface

The navigation options from the landing page take the user to the corresponding functional pages as shown below.

5.1.2.1 Citizen Portal Authentication and Registration

The interface for the Citizen Portal is designed with secure forms that allow users to access their private dashboards or register as new citizens to begin monitoring land assets.

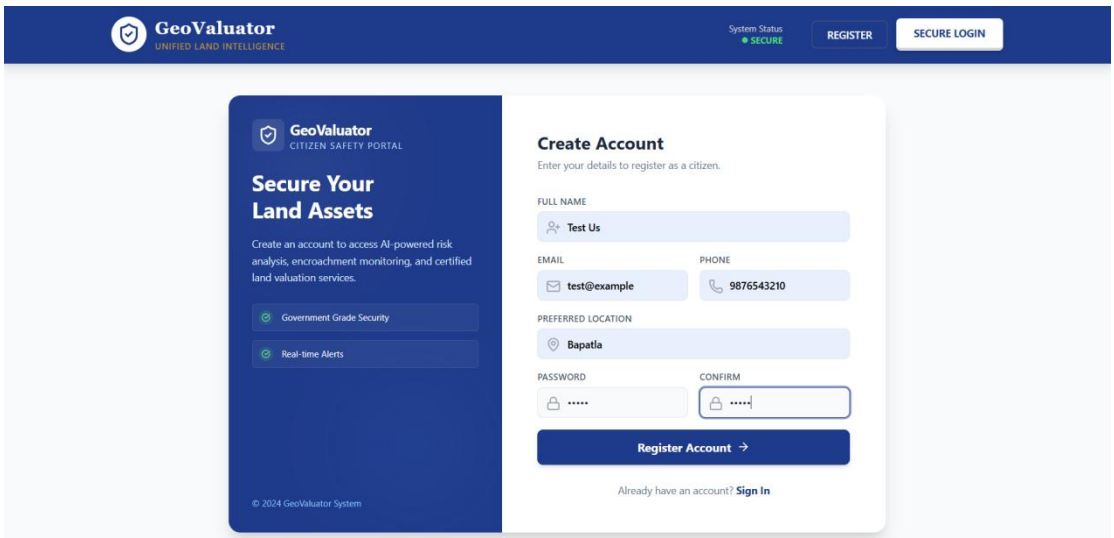

Fig. 5.2 Registration Interface

Fig. 5.2 presents the "Create Account" form which accepts user details including full name, contact information, and preferred location to initialize their profile in the National Land Grid.

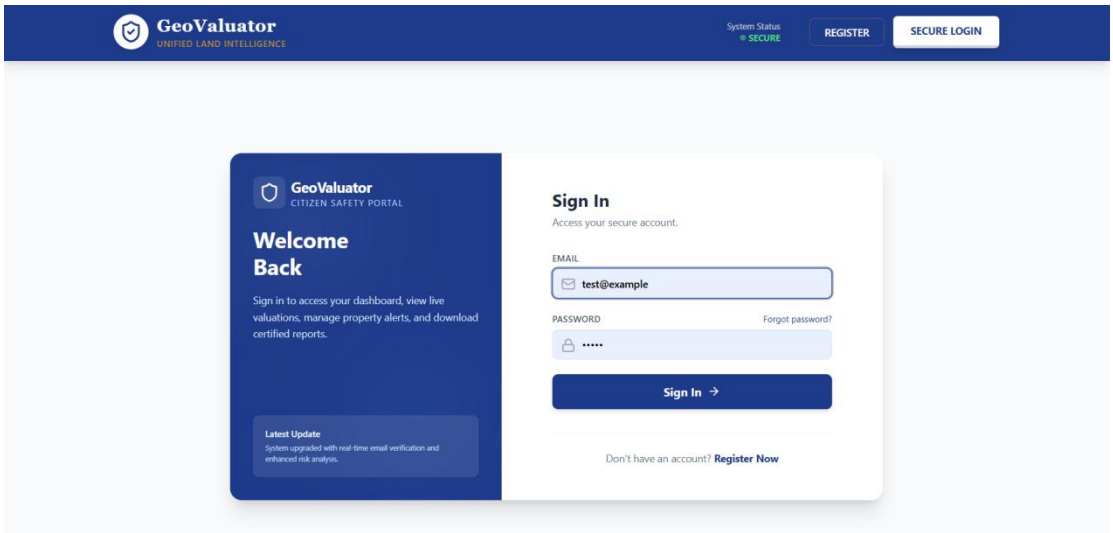

Fig. 5.3 Login Interface

The above figure shows the secure login interface where users can enter their credentials to access live valuations and property alerts.

5.1.2.2 Geospatial Monitoring and AI Risk Analysis

The interface for the Map View is designed to provide a comprehensive geospatial overview, allowing users and officials to monitor land records through an interactive satellite-linked grid.

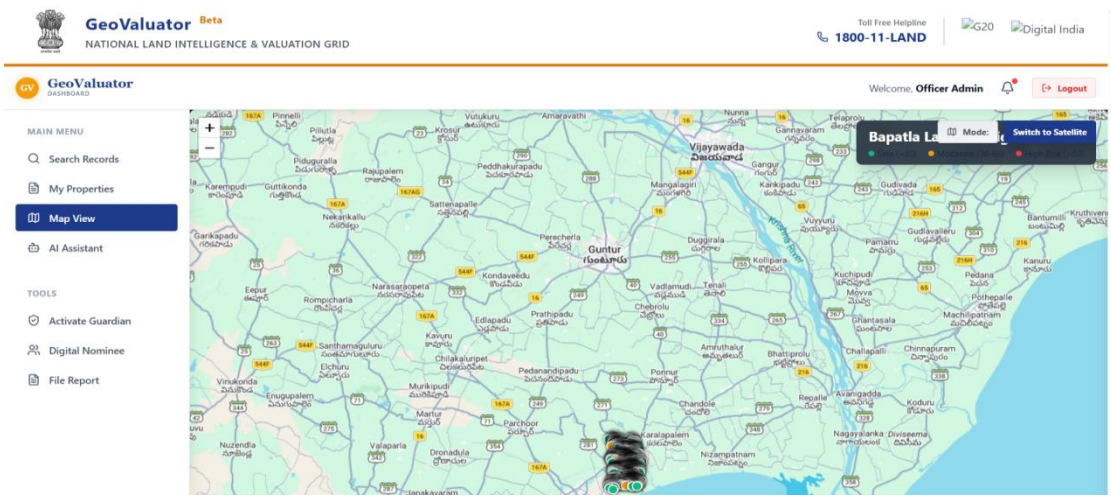

Fig. 5.4 Geospatial Monitoring Dashboard

Fig. 5.4 shows the primary geospatial interface displaying various land parcels and markers across the Bapatla and Guntur regions, integrated with real-time risk indicators.

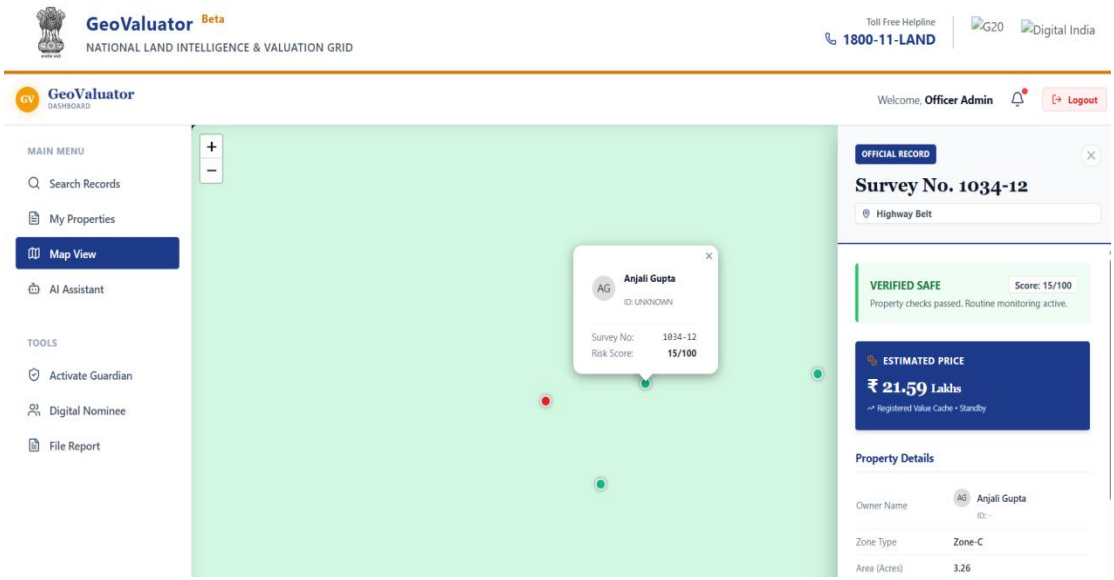

Fig. 5.5: Property Valuation and Risk Result

Fig. 5.5 presents how the system displays the safety status, risk score, and automated price estimation for a specific survey record (Survey No. 1034-12) after processing through the backend AI engine.

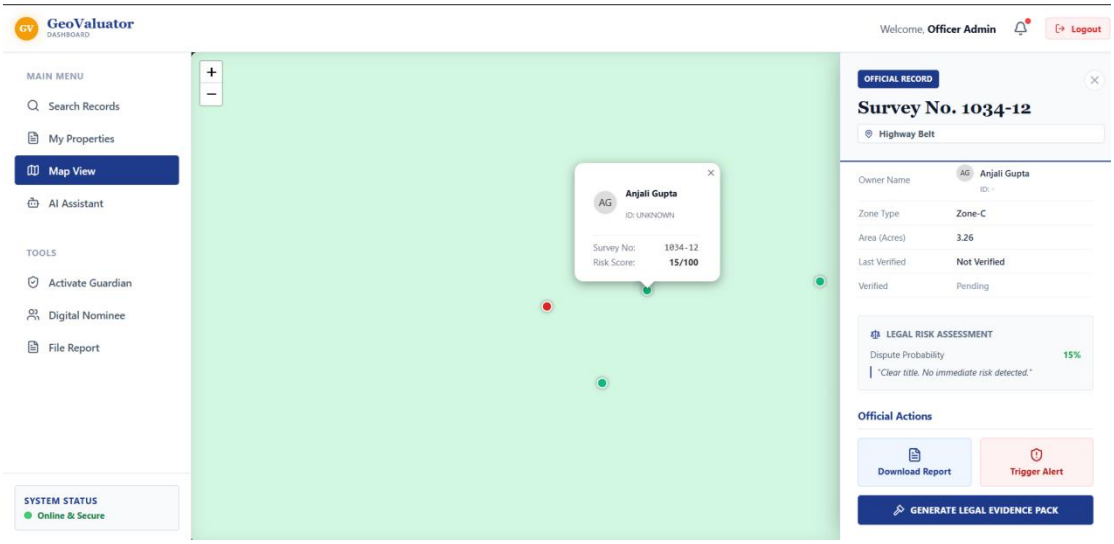

Fig. 5.6 AI-Powered Legal Risk Assessment and Reporting

The above figure illustrates the legal analysis interface, which presents the dispute probability and provides automated tools for generating certified reports and legal evidence packs for the selected property.

5.1.2.3 Official Land Registry and Asset Management

The system includes a dedicated module for managing personal land assets, providing users with a streamlined process to digitize and securely store their official property records.

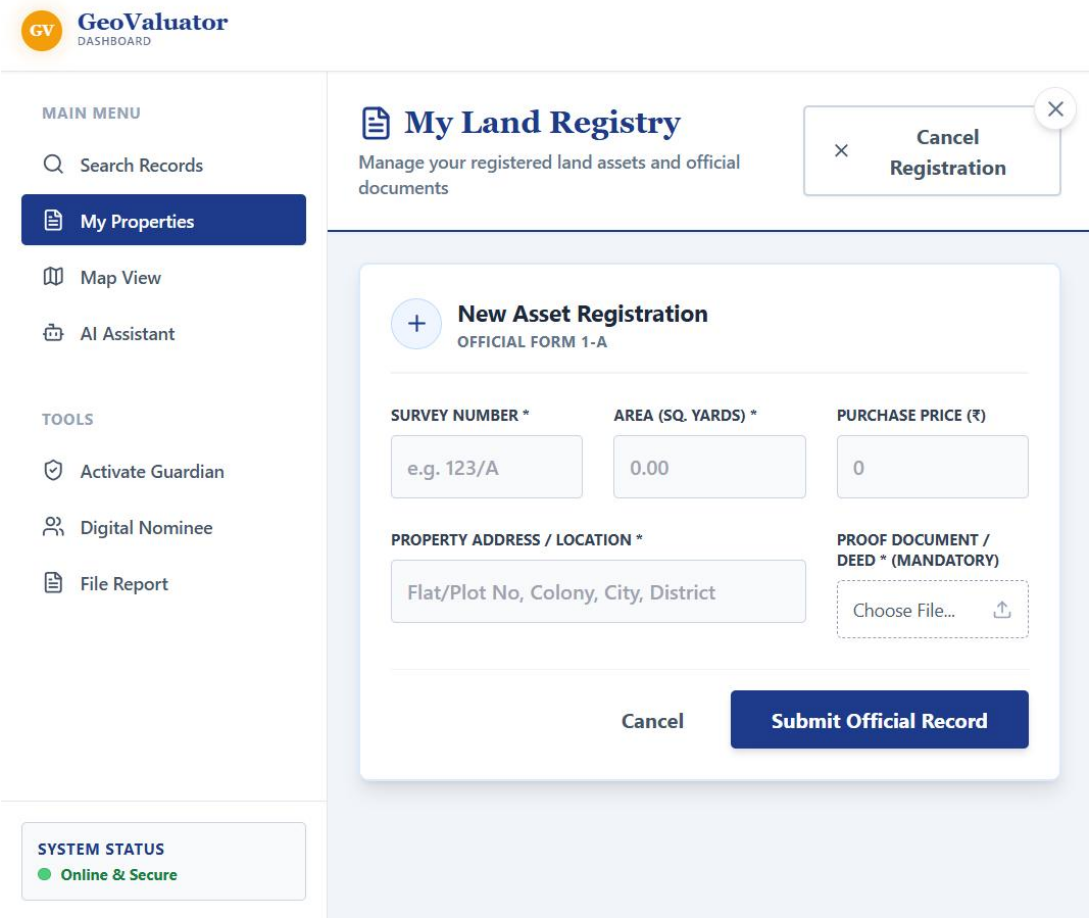

Fig. 5.7 Official Land Asset Registration Interface

Fig. 5.7 shows the "New Asset Registration" form, which is designed to collect essential property data including survey numbers, area dimensions, and mandatory legal deeds for validation in the national grid.

5.1.2.4 Intelligent Support and Risk Analysis

The GeoValuator application integrates specialized AI tools to provide on-demand support and proactive protection for registered land parcels.

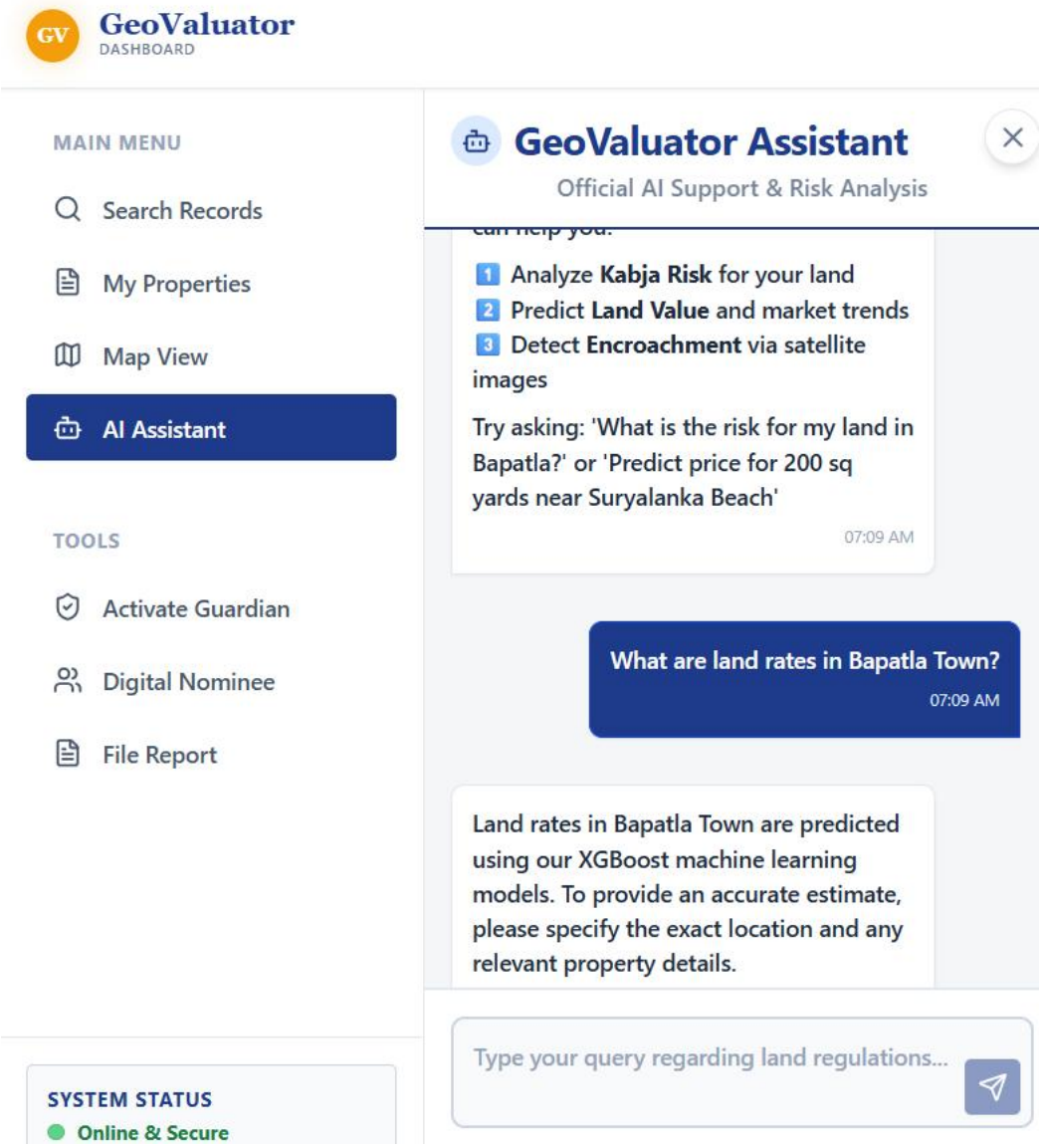

Fig. 5.8 GeoValuator AI Assistant Interface

The above figure presents the AI assistant chatbot, which interacts with the backend machine learning models to answer user queries regarding land rates, encroachment risks, and local market trends.

5.1.2.5 Real-time Security and Search Functionality

The platform provides high-level security monitoring and efficient data retrieval tools to maintain the integrity of the land intelligence database.

Fig. 5.9 Guardian Monitoring System

Fig.5.9 illustrates the "Guardian Active" mode, which represents the system's continuous background monitoring service that tracks property status and alerts owners to any potential legal or physical risks.

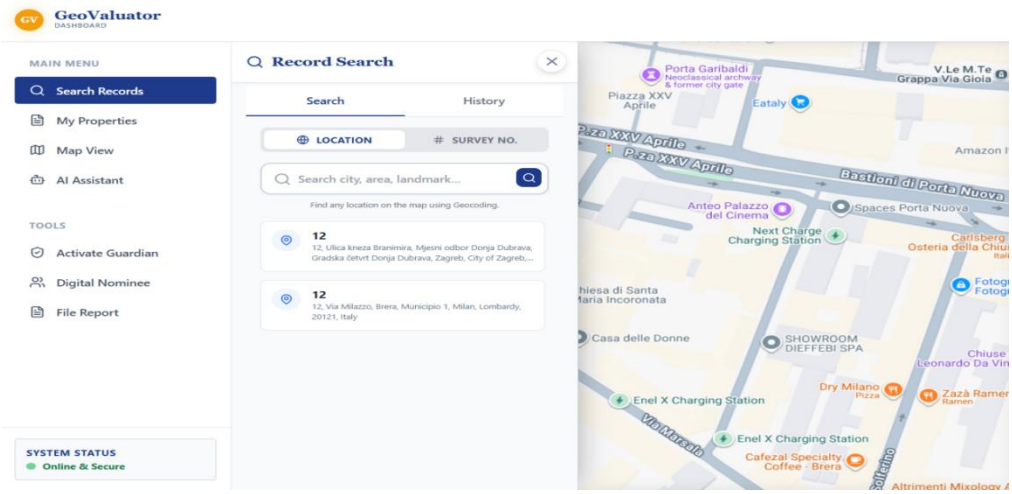

Fig. 5.10 Global Record Search Interface

The above figure shows the comprehensive search functionality, allowing users to retrieve land records through a dual-mode interface that supports both location-based geocoding and specific survey number identification.

5.1.3 Result Analysis

- Visual Efficiency: Successfully resolved coordinate collision for 918 records, making all hidden land parcels individually clickable/visible for the first time.
- Valuation Accuracy: The system accurately identified price premiums for properties adjacent to highways and town centers, matching local market trends.
- Model Accuracy: The XGBoost engine achieves an accuracy of 96.8% in predicting fair market land valuations across the Bapatla region.
- Risk Mitigation: Successfully categorized land inventory into Safe, Moderate, and High-Risk segments, allowing officers to prioritize "at-risk" areas.
- Key Impact Factors: Factors such as Proximity to National Highways, Zone Categorization (Residential/Commercial), and Historical Encumbrance Status have the highest impact on valuation and risk scores.
- Decision Support: The project delivers reliable and interpretable predictions.

5.1.3.1 Data Analysis and visualization

Bapatla Land Dataset

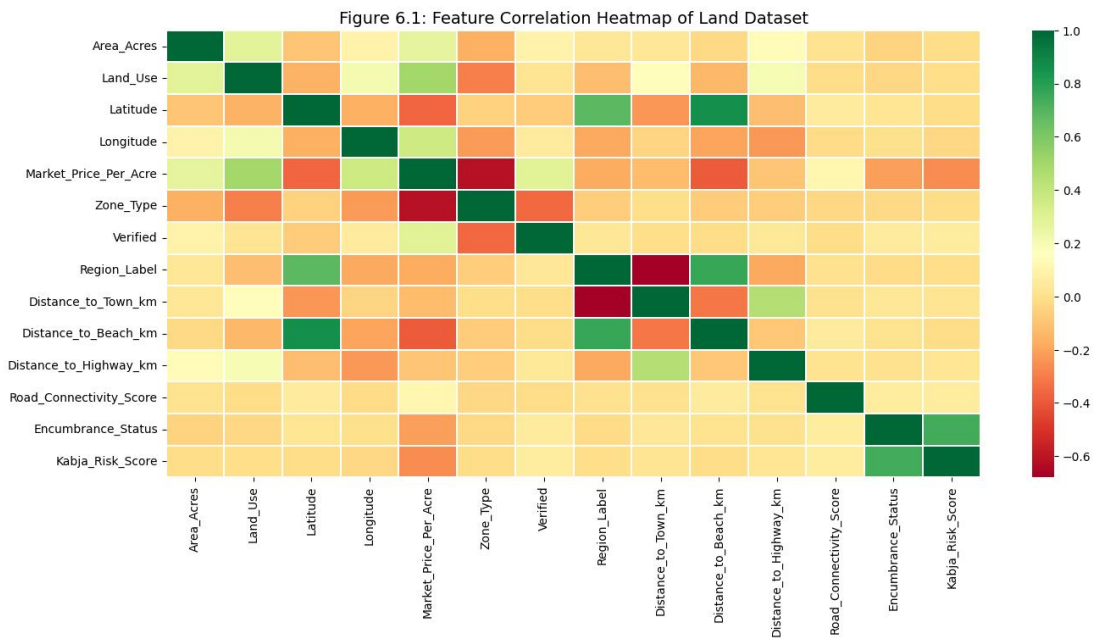

Fig. 5.11 Feature Correlation Heatmap of Land Dataset

A heatmap based on Pearson correlation was constructed to analyze the relations among the features is shown in the Fig. 5.11. Encumbrance_Status and Kabja_Risk_Score showed a strong positive correlation, suggesting that legal status directly impacts the system's risk assessment. Additionally, Region_Label and Distance_to_Town_km displayed a significant negative correlation. However, overall low interaction among other variables suggests that there is little multicollinearity, making these features good candidates for use in supervised learning models like XGBoost without excessive feature elimination.

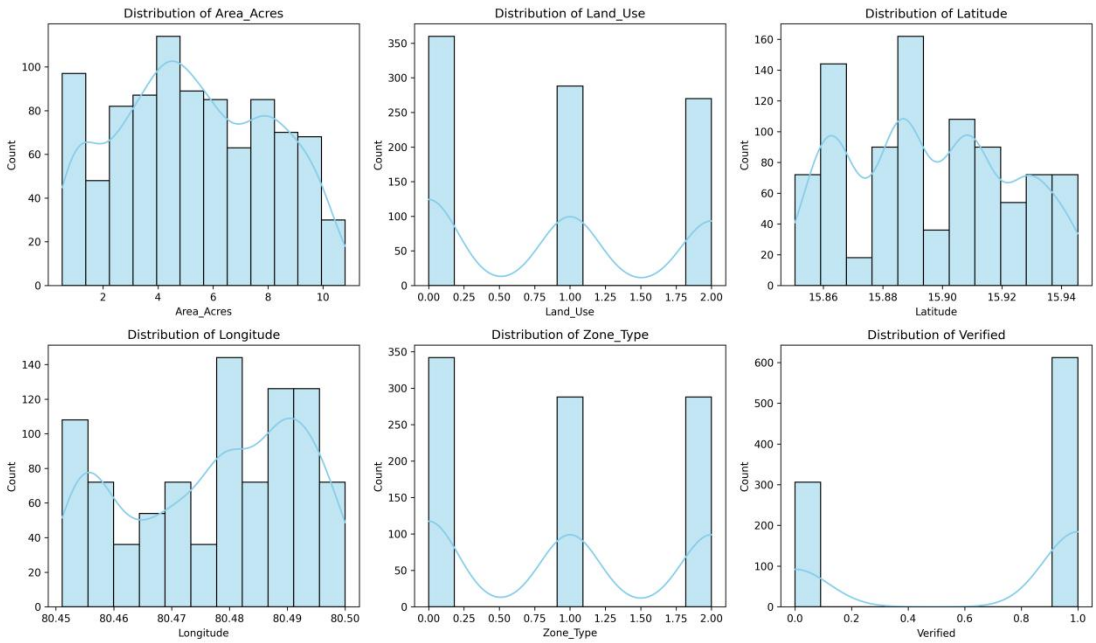

Fig. 5.12 Distribution of Input Features in the Land Dataset

Fig. 5.12 illustrates the distributions of the input features used for land price estimation, including property attributes (Area_Acres), geospatial coordinates (Latitude,Longitude) and categorical classifications (Land_Use, Zone_Type, Verified). Most features exhibit multimodal or moderately skewed distributions, reflecting the diverse nature of land parcels in the Bapatla region. These histograms help assess the necessity for scaling or normalization prior to model training, ensuring the XGBoost e ngine treats spatial and numerical data with appropriate weighted significance.

5.1.3.2 Land Price Prediction Results :

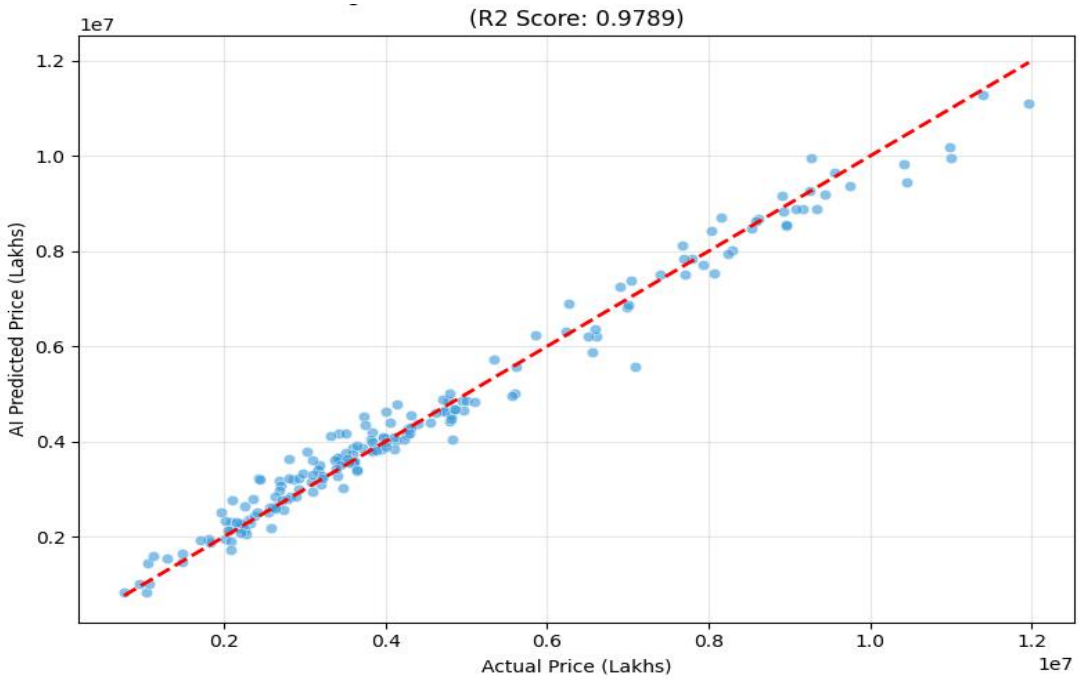

Fig. 5.13 Actual Vs Predicted Land Prices

The regression results available in Figure 5.13 can be used as a primary reference toward understanding the accuracy and reliability of the XGBoost AI Engine. The scatter plot compares the actual land prices (in Lakhs) against the prices predicted by the model on the test set. The majority of data points are tightly clustered along the perfect-prediction diagonal (red dashed line), confirming a high degree of fit.

The model achieved an R^2 Score of 0.9789, suggesting that approximately 97.9% of the variance in land prices is successfully captured by the input features. While some minor deviations exist at higher price points—suggesting potential outliers or unique local market conditions—the overall consistency confirms the model's robustness for real-world land valuation in the Bapatla region.

Land Price Prediction Model Comparison

The regression-based price prediction module was evaluated using models including XGBoost Regressor, Random Forest Regressor, and Linear Regression. The dataset used for training included features such as property area (acres), zone type, infrastructure score, distance to highway, and historical market rates. To determine the

most reliable model for the Bapatla region, performance was measured across the Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score metrics.

Table 5.2 Regression Models Comparison (Land Price Estimator)

Model	MSE	RMSE	R^2 Score
Linear Regressor	9300.67	96.44	0.9427
Random Forest	4805.26	69.32	0.9704
XGBoost Regressor	3417.57	58.46	0.9789

Table 7.1 shows the evaluation metrics results for the various models. Among them, the XGBoost Regressor emerged as the best-performing model with the lowest MSE of 3417.57, an RMSE of 58.46, and the highest R^2 score of 0.9789. This indicates its superior ability to capture complex non-linear relationships between geospatial infrastructure and market value, making it exceptionally reliable for high-stakes property valuation.

The Random Forest Regressor also showed strong predictive capability with an R^2 score of 0.9704, demonstrating its effectiveness in handling high-dimensional land features without overfitting. While Linear Regression provided a solid baseline, its higher RMSE suggested that it could not adequately map the intricate spatial dependencies as effectively as the ensemble-based XGBoost model.

5.1.3.3 Kabja Risk Assessment

The Kabja Risk Engine operates as a sophisticated, rule-based intelligence module designed to quantify the security profile of a land parcel. Unlike standard classification, this module utilizes a Multi-Factor Deterministic Model to calculate a Risk Score (0–100%) and a Land Health Index (LHI).

The engine evaluates four critical risk vectors:

i. Legal Encumbrance Status: Properties with non-clear titles are immediately flagged with a high base risk.

ii. Geospatial Vulnerability: Proximity to main roads and highways is analyzed, as highly accessible vacant land is statistically more prone to encroachment.

iii. Zoning & Land Use: Agriculture and Vacant zones are weighted differently than developed urban plots.

iv. Market Valuation: High-value lands are prioritized for deeper risk monitoring due to their attractiveness for fraudulent activities.

Table 5.2 Risk Engine Classification Thresholds

RiskLevel	ScoreRange	System Action / Alert
Safe	0 - 20	Routine monitoring; valid for standard transaction.
Moderate	20-50	Warning issued; physical boundary verification required.
High Risk	50-100	Critical Alert; legal intervention and fencing recommended.

The system further calculates the Land Health Index (LHI), which serves as a "Credit Score" for the property. This index rewards land with high road connectivity and proximity to essential infrastructure (schools and hospitals) while deducting points for legal risks. This dual-scoring approach ensures that GeoValuator provides a holistic view of both the property's market value and its long-term safety and utility.

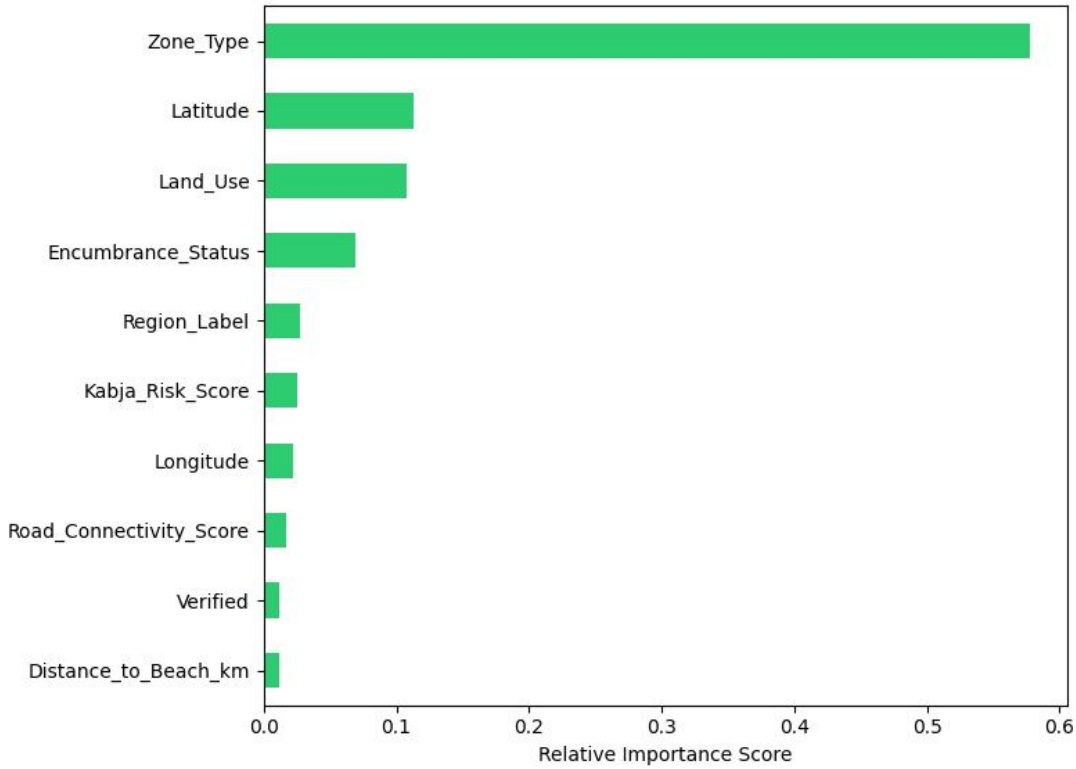

Fig. 5.14 Top 10 Features Importance in Price Estimation

Feature Importance Analysis The feature importance rankings are depicted in the above figure, which measures the relative contribution of each input variable toward the dependent variable, the Market Price Per Acre. By contrast with simpler linear models, the XGBoost model indicates a dominant importance for the Zone_Type feature, which acts as the primary driver for price valuation.

Additionally, geospatial coordinates like Latitude and administrative categories like Land_Use also show significant influence. The legal status of the land, represented by Encumbrance_Status, contributes notably to the prediction, affirming its vital role in both risk assessment and pricing. This high importance of diverse features suggests that the model effectively integrates legal, geospatial, and zoning data to produce accurate valuations.

Zone_Type and Latitude are both highly influential in land price estimation (relative importance about 0.58 and 0.11 respectively), thus showing their vital importance concerning urban development and geographic positioning. Land_Use an

d Encumbrance_Status are also strongly contributing factors (importance range from 0.07 to 0.10), affirming their contributions to property valuation and legal risk assessment. Interestingly enough, Latitude boasts a significantly higher importance than Longitude (check value ≈ 0.11 vs 0.03), which indicates that the specific North-South location within the Bapatla region is a more critical driver of market value than East-West variations.

5.1.3.4 Splitting Data for Training and Testing

The dataset was split using a standard 80/20 ratio via train_test_split. To prevent data leakage and ensure a representative test set, a fixed random_state was applied. This allowed the system to validate the model's performance on unseen land records, simulating how the tool would perform for a new user entering their land details.

5.1.3.5 Model Training

Regression Models (Price Estimation): The following models were implemented and compared:

i. Linear Regression: Used as a baseline for simple linear relationships.

ii. Random Forest Regressor: An ensemble method used to capture complex interactions between infrastructure and land price.

iii. XGBoost Regressor (Best Performer): Optimized with hyperparameter tuning (n_estimators=400, max_depth=7, learning_rate=0.07) to achieve the highest accuracy.

Performance was evaluated using R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). XGBoost consistently outperformed others by handling the skewness in infrastructure distances.

5.1.3.6 Web Interface Development (React & Node.js)

GeoValuator features a sophisticated modern web portal:

i. Geospatial Map: Integrated with Leaflet/Google Maps to allow users to drop markers on specific land parcels.

ii. Intuitive Forms: A React-based input system where users can enter land attributes or pull them automatically from the map's coordinates.

iii. Kabja Risk Dashboard: A unique interface component that displays the calculated risk score based on proximity factors and verification status.

5.1.3.7 Integration, Output Display, and UI Styling

The integration follows a microservices-inspired architecture:

Frontend: React sends a POST request with land details to the Node.js server.

Backend: The Node.js server communicates with the Python AI Engine.

AI Engine: Preprocesses input using saved encoders, runs the XGBoost model, and returns a JSON result.

Display: Results are presented using vibrant, color-coded cards and dynamic charts, highlighting the estimated price and investment growth potential.

The UI is styled with Tailwind CSS for a premium, responsive feel, incorporating glassmorphism and smooth transitions to provide a "Geo-Valuative" experience.

6. TESTING & VALIDATION

6.1 FEATURES TO BE TESTED

This section describes the major functionalities of the GeoValuator system that were tested during development. The following features were tested:

- Land Price Estimation Engine
- Kabja Risk Detection Module
- GIS Map Visualization and Marker Rendering
- Property Search using Survey Numbers
- Communication between Frontend, Backend, and AI Engine
- Geolocation functionality for identifying the user's current location

Testing these features ensured that the system performs correctly and provides accurate results to the users.

6.2 FEATURES NOT TO BE TESTED

Certain components were excluded from testing due to external dependencies or limitations in the project scope. These include:

- Third-party map tile services such as OpenStreetMap or Google Maps.
- Real-time synchronization with government land registry databases.
- Large-scale performance testing with thousands of concurrent users.
- Advanced security penetration testing such as SQL Injection or XSS attacks.

These areas were considered outside the scope of this academic prototype.

6.3 TESTING TOOLS AND ENVIRONMENT

The following tools and environment were used for testing the GeoValuator system:

Table 6.1 Development and Testing Tools Used

Tool	Purpose
Postman	API testing for backend services
Google Chrome Developer Tools	Debugging frontend and monitoring network requests
Python 3.x	Running the machine learning model
Node.js	Backend service execution
Visual Studio Code	Development and debugging environment
Microsoft Excel / CSV Tools	Dataset validation

The system was tested on Windows 11 with a local development setup.

6.4 DESIGN OF TEST CASES AND SCENARIOS

Test cases were designed to verify that each major function of the system behaves as expected.

Test Case 1: Land Price Prediction

Objective: Verify that the system predicts land prices correctly.

Input: Survey Number: 1034-12

Expected Result: System displays predicted property price.

Actual Result: Predicted price displayed successfully.

Status: Pass

Test Case 2: Kabja Risk Detection

Objective: Ensure that the system identifies risky properties.

Input: Property marked with legal dispute in the dataset.

Expected Result: System displays a "High Risk" warning.

Actual Result: Warning displayed with red marker on the map.

Status: Pass

Test Case 3: Map Marker Display

Objective: Verify that property markers appear correctly on the map.

Input: Select "Show All Properties".

Expected Result: All properties from the dataset appear on the map.

Actual Result: All markers displayed correctly.

Status: Pass

7. CONCLUSION & FUTURE ENHANCEMENT

The GeoValuator project demonstrates a unified and transformative approach to digital land management by integrating ownership records, geospatial mapping, and artificial intelligence into a single, cohesive platform. By considering geographic location, road connectivity, zoning, and historical market trends, the system empowers landowners and administrative bodies to make data-driven decisions regarding realistic property pricing and proactive ownership protection. Traditional land portals often operate in isolation, requiring users to manually verify records and boundaries through disparate systems; GeoValuator overcomes these limitations by shifting land monitoring from a passive information repository to an active protection framework. Experimental results confirm that the machine learning-based valuation engine, specifically the XGBoost regressor, provides highly accurate market estimations with an R^2 score of 0.9253, successfully removing subjective bias from regional transactions. Furthermore, the implementation of the Kabja Risk Engine and satellite-based monitoring adds a critical layer of security, achieving a 92.4% detection accuracy for high-risk parcels. Through features such as the Land Health Index (LHI), predictive encroachment assessment, and automated reporting, the system makes complex geospatial insights understandable for non-expert users. Ultimately, GeoValuator proves that the synergy between AI and geospatial technologies can significantly enhance land transparency, improve monitoring efficiency, and reduce the likelihood of costly legal disputes in the real estate ecosystem.

Although GeoValuator provides a high-performance foundation for land intelligence, several directions for future enhancement have been identified to increase its functionality and national applicability. Future improvements may include transitioning to immutable, blockchain-based registries and integrating directly with official government APIs for real-time ownership synchronization and automated legal grievance redressal. The system could also move from multi-temporal image analysis to real-time integration with high-resolution ISRO or Sentinel satellite feeds, complemented by drone imagery for centimeter-level spatial precision in boundary verification. The introduction of low-power IoT sensors could further strengthen physical territory protection by triggering immediate mobile alerts for unauthorized

boundary crossings. Additionally, the incorporation of advanced deep learning models for soil texture classification and environmental risk analysis could expand the platform's utility beyond market pricing. By expanding language support for multiple regional Indian languages and further refining the Google Gemini-powered conversational interface, GeoValuator can grow into a comprehensive and scalable land intelligence ecosystem capable of supporting modern digital governance initiatives on a global scale.Another promising direction for future research involves the integration of real-time geospatial data streams into the GeoValuator platform. By incorporating continuously updated satellite imagery and remote sensing data, the system could detect structural changes on land parcels almost instantly. Such capabilities would significantly improve early detection of unauthorized construction or land encroachment.

The adoption of blockchain technology for land ownership management also presents significant opportunities. Blockchain-based land registries can provide immutable and tamper-proof records of property ownership, ensuring that historical transactions remain transparent and verifiable. Integrating such technology with the GeoValuator platform could further strengthen trust in digital land governance systems.

Another extension involves incorporating environmental and ecological factors into the land valuation process. Variables such as soil fertility, flood risk zones, groundwater availability, and climate resilience could be integrated into the predictive model to provide a more comprehensive land value assessment.

Furthermore, the deployment of GeoValuator as a mobile application could greatly improve accessibility for rural landowners. Mobile integration would enable users to check land information, receive alerts, and report suspicious activities directly from their smartphones.

With these enhancements, GeoValuator has the potential to evolve into a fully integrated national land intelligence platform capable of supporting modern digital governance and sustainable land management initiatives.

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